

Week 5 Assignment

Data Warehousing & Business Intelligence using Power BI

A practical, industry-aligned handbook designed to master modern enterprise data warehousing concepts, relational star schema design, ETL pipelines, DAX measure creation, and executive dashboard development using Microsoft Power BI.

OLTP Database

ERP / CRM

CSV / Excel

STUDENT NAME: Maira Naveed

1. LEARNING OUTCOMES & INDUSTRY CONTEXT

Program Objectives: This practical assignment is designed to bridge academic theoretical concepts in database systems with enterprise-grade Data Warehousing (DW) and Business Intelligence (BI) techniques. By completing this project, students simulate the role of an Enterprise BI Developer working in an agile environment.

Key Competencies Developed:

- **Data Warehouse Foundations:** Understand analytical architecture vs transactional processing.
- **Dimensional Data Modeling:** Identify and categorize Fact and Dimension tables and determine keys.
- **Schema Architecture:** Build, normalize, and visualize high-performance Star Schemas.
- **Power Query & ETL:** Ingest, cleanse, and transform tabular enterprise datasets into Power BI.
- **Relationship Modeling:** Establish primary/foreign key relationships with cardinality and filter propagation.
- **DAX Engineering:** Author custom Data Analysis Expressions for KPIs and dynamic aggregations.
- **Executive Dashboard UI/UX:** Design publication-ready dashboards formatted for executive leadership.
- **Business Decision Intelligence:** Extract actionable insights and formulate recommendations for C-suite executives.

2. ENTERPRISE CASE STUDY: ABC RETAIL STORE

Business Scenario:

ABC Retail Store is a nationwide omnichannel retail enterprise operating across multiple states. The company specializes in three major product categories: **Electronics, Furniture, and Office Supplies**.

Currently, executive decision-makers suffer from operational visibility bottlenecks due to data fragmentation across siloed operational transactional systems. The Chief Executive Officer (CEO) and Chief Commercial Officer (CCO) require an interactive, real-time Business Intelligence Executive Dashboard in Microsoft Power BI to instantly assess revenue patterns, geographic performance, customer engagement, and product profitability.

As a newly appointed **Data Engineer / BI Developer**, your mandate is to ingest raw transactional data, transform it into a dimensional model (Star Schema), write core analytical DAX measures, and present a dynamic, interactive dashboard.

Provided Datasets Overview:

The enterprise operational team has exported three core relational tables in Microsoft Excel format:

1. Sales Dataset

Contains operational transaction log data including Order Identifiers, Order Dates, Customer Identifiers, Product Identifiers, Quantities, and Revenue amount.

2. Customers Dataset

Contains descriptive attributes regarding the retail client base, including unique Customer Identifiers, Full Names, and Geographic Cities.

3. Products Dataset

Contains detailed master item metadata including Product Identifiers, Descriptive Names, and High-Level Category classifications.

3. DATASET SPECIFICATIONS & TABULAR SAMPLES

Below are the explicit tabular samples of the datasets provided by ABC Retail Store for model construction.

Sales Table (Transactional Sample)

OrderID	Date	CustomerID	ProductID	Quantity	Sales (\$)
ORD-1001	2026-01-05	CUST-101	PROD-501	2	1,200
ORD-1002	2026-01-06	CUST-102	PROD-503	1	450
ORD-1003	2026-01-10	CUST-103	PROD-502	5	150
ORD-1004	2026-01-15	CUST-101	PROD-504	3	900
ORD-1005	2026-01-18	CUST-104	PROD-501	1	600
ORD-1006	2026-01-22	CUST-105	PROD-505	4	200
ORD-1007	2026-02-01	CUST-102	PROD-502	10	300
ORD-1008	2026-02-05	CUST-103	PROD-501	2	1,200
ORD-1009	2026-02-12	CUST-104	PROD-503	2	900
ORD-1010	2026-02-20	CUST-105	PROD-504	1	300

Customers Table (Master Data)

CustomerID	Customer Name	City
CUST-101	Ali Khan	Lahore
CUST-102	Sara Ahmed	Karachi
CUST-103	Usman Malik	Islamabad
CUST-104	Ayesha Raza	Lahore
CUST-105	Bilal Sheikh	Peshawar

Products Table (Master Data)

ProductID	Product Name	Category
PROD-501	Laptop Ultra	Electronics
PROD-502	Ergonomic Desk	Furniture
PROD-503	Smart Monitor	Electronics
PROD-504	Executive Chair	Furniture
PROD-505	Paper Shredder	Office Supplies

Table Classification Explanation:

Fact Table (Sales): Contains numeric, measurable metrics (Quantity, Sales) that reflect discrete business events or transactions. It maintains high volatility and grows continuously as sales occur.

Dimension Tables (Customers & Products): Contain contextual, descriptive background attributes (Name, City, Category) used to filter, slice, group, and categorize the numerical measures stored in the Fact Table.

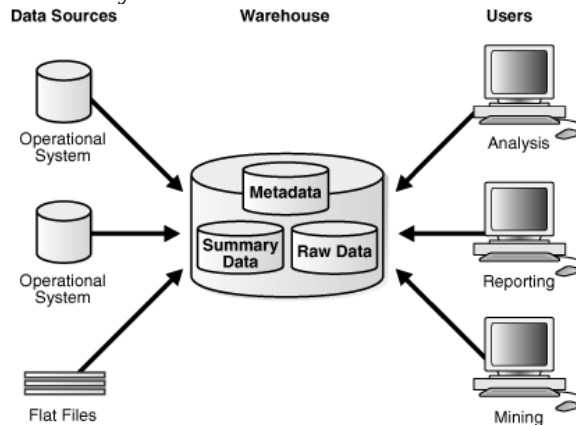
TASK 1: FUNDAMENTALS OF DATA WAREHOUSING

5 MARKS

Instructions: Provide structured, technically precise answers to the following foundational data engineering questions.

1.1 What is a Data Warehouse?

A data warehouse is a centralized repository designed to aggregate, store, and manage structured historical data from multiple sources to support business intelligence (BI), reporting, and strategic decision-making. It's not used for daily operations — it's used so people can look back at old data and create reports and analyze trends easily.



1.2 Why do enterprises deploy Data Warehouses instead of running analytical queries on Operational

Operational databases are built for quick daily tasks like adding a new order or updating a customer's info — not for big, heavy reports. If you run large reports directly on them, it slows down the actual business (like the checkout system). A Data Warehouse keeps reporting separate from daily operations, so both work faster and don't interfere with each other.

1.3 Architectural Comparison: OLTP vs. Data Warehouse (OLAP)

Dimension / Feature	OLTP (Operational Database)	OLAP / Data Warehouse
Primary Purpose	Day-to-day transactional operations	Analytical reporting & decision making
Data Structure	Highly Normalized (3NF / BCNF)	Denormalized (Star / Snowflake Schema)
Query Characteristics	Fast, simple reads/writes (INSERT/UPDATE)	Complex analytical aggregations (SUM/AVG)
Data Horizon	Current operational data (Short-term)	Historical multi-year trend data

1.4 Describe the Enterprise ETL (Extract, Transform, Load) Process:

Extract – Take the raw data from the original files (Sales, Customers, Products Excel sheets).
Transform – Clean it up: fix wrong formats, remove duplicates, fill or handle missing values, and make sure numbers/dates are stored correctly.
Load – Put the cleaned data into Power BI so it's ready to use for reports and dashboards.

TASK 2: FACT & DIMENSION TABLE ANALYSIS

10 MARKS

Instructions: Analyze the provided ABC Retail Store dataset schemas and complete the technical classification matrix below.

2.1 Schema Classification Table

Table Name	Table Classification (Fact / Dimension)	Primary Key (PK)	Foreign Keys (FK)
Sales	Fact Table	OrderID	CustomerID, ProductID
Customers	Dimension Table	CustomerID	None
Products	Dimension Table	ProductID	None

2.2 Measure & Attribute Identification

Categorize all fields across the three tables into **Quantitative Measures** (numeric, additive data) and **Descriptive Attributes** (categorical text/dates).

Quantitative Measures:

Quantity, Sales (\$)

Descriptive Attributes:

OrderID, Date, CustomerID, ProductID, Customer Name, City, Product Name, Category

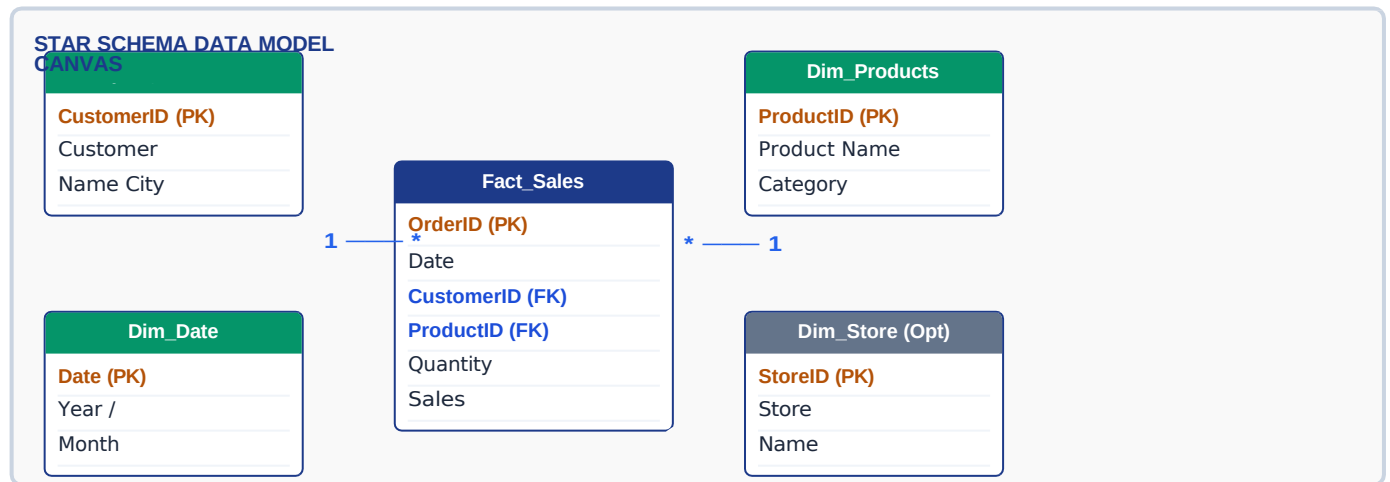
2.3 Importance of Surrogate Keys in Data Warehousing

A surrogate key is a simple, made-up ID number (like 1, 2, 3...) created by the Data Warehouse itself, instead of using the ID from the original system (like CUST-101). It's better because the original IDs can change, get reused, or be different across systems, but a surrogate key always stays the same and doesn't depend on outside systems. This keeps the data warehouse stable and consistent even if source systems change.

TASK 3: STAR SCHEMA DATA ARCHITECTURE

10 MARKS

Instructions: Diagram the complete Star Schema model for ABC Retail Store. Show the central Fact Sales table surrounded by Dimension tables (Customers, Products, Date Dimension, and optional Store Dimension). Indicate Primary Keys (PK), Foreign Keys (FK), and relationship cardinalities (1:*).



Student Task: Review the Star Schema visual above. Verify that all 1-to-Many (1:*) cardinalities flow correctly from Dimension tables (1 side) to the Fact table (* side). Write a brief architectural summary below justifying why Star Schema provides faster query speeds than normalized schemas.

A Star Schema is faster than a normal normalized database because it has fewer tables and fewer "joins" needed to connect them. In this design, the Fact_Sales table sits in the middle and connects directly to each dimension table (Customers, Products, Date) with a simple one-step link. This means when you run a report — like "total sales by city" — the system doesn't have to jump through many linked tables to get the answer. Fewer joins means quicker queries, which is exactly what's needed for fast, interactive dashboards used by executives.

TASK 4: POWER BI INGESTION & POWER QUERY

10 MARKS

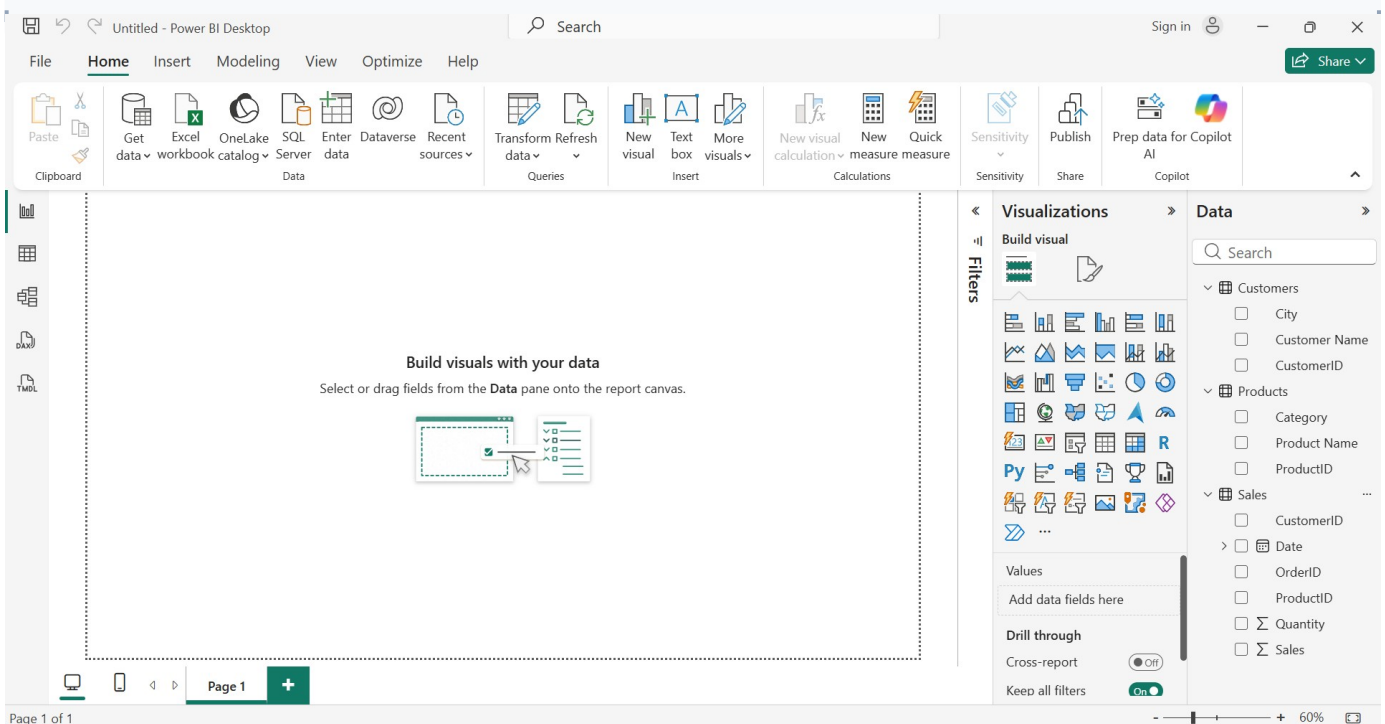
Step-by-Step Instructions:

1. Launch **Power BI Desktop**. Click *Get Data* → *Excel Workbook* and select the source file.
2. Select all three sheets: Sales, Customers, and Products.
3. Click **Transform Data** to open Power Query Editor. Ensure correct data types (Dates as Date, Sales as Decimal, Quantity as Integer).
4. Click **Close & Apply**.



[SCREENSHOT PLACEHOLDER: POWER QUERY EDITOR / LOADED TABLES]

Paste a screenshot showing the Power BI Data Pane containing Sales, Customers, and Products tables loaded.

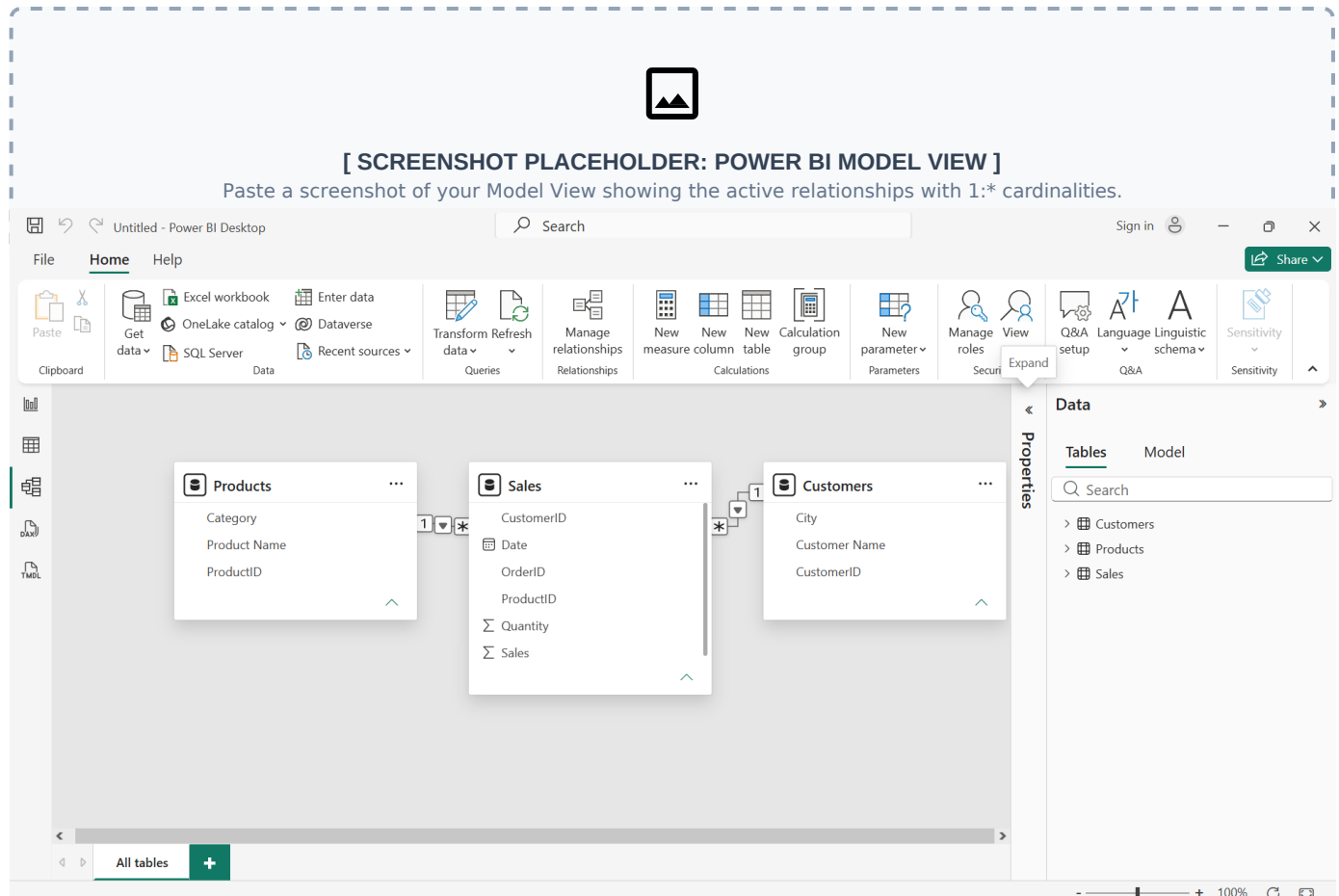


TASK 5: ESTABLISHING DATA RELATIONSHIPS

10 MARKS

Navigate to **Model View** in Power BI. Create active relationships between tables:

- Customers.CustomerID (1) — filter propagation —> Sales.CustomerID (*)
- Products.ProductID (1) — filter propagation —> Sales.ProductID (*)



Technical Explanation: Why are relationship directions and single-cross filtering critical?

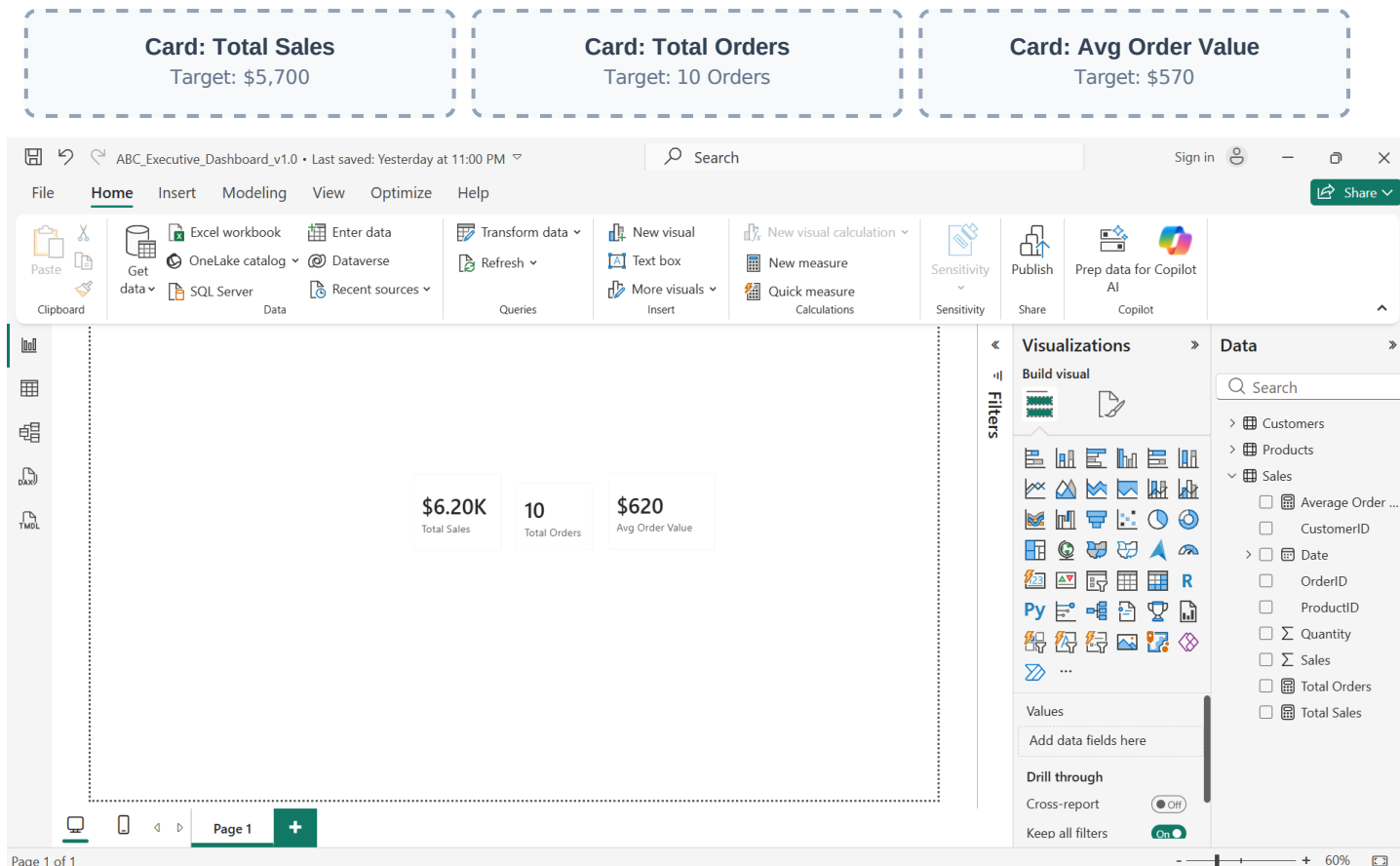
Relationship direction controls which table can filter the other. In this model, Customers and Products (the "1" side) filter Sales (the "many" side) — so selecting a city filters down to only that city's sales. Single-direction cross filtering keeps this flow predictable and prevents circular filtering, which could confuse results or slow down performance. Bi-directional filtering is avoided unless specifically needed, since it can cause ambiguous filter paths across multiple tables.

TASK 6: EXECUTIVE DASHBOARD VISUALS

30 MARKS

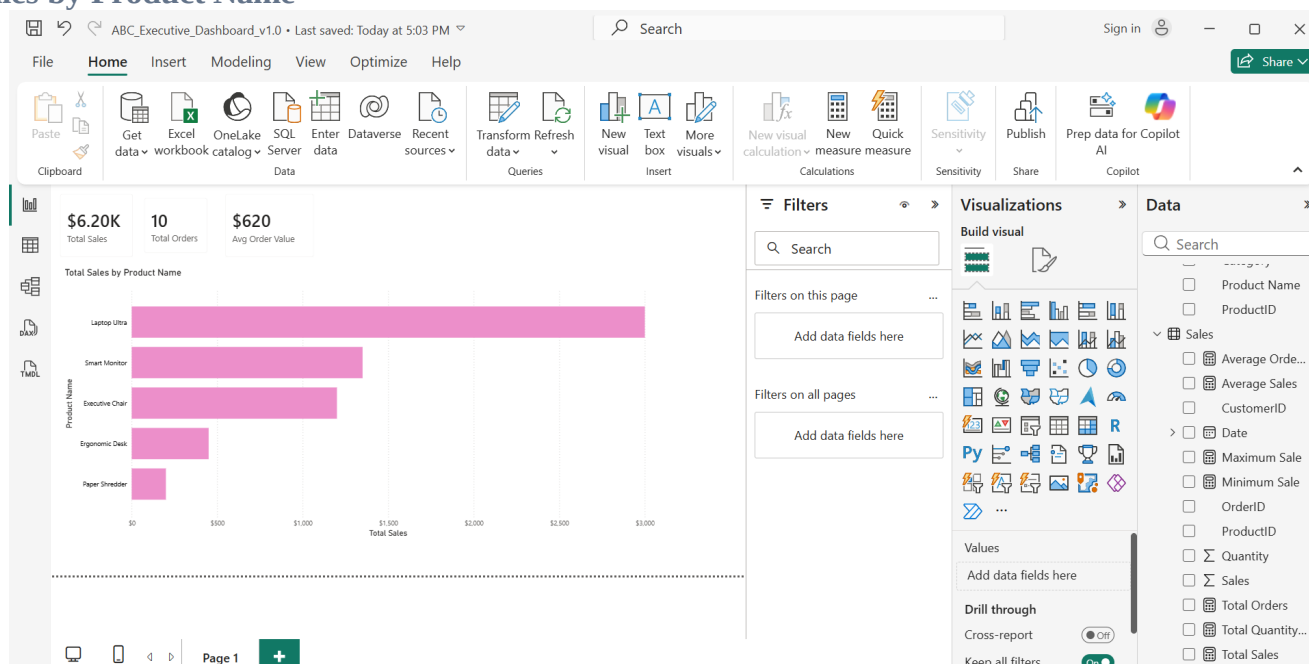
Create the specified visuals in Power BI Report View. Place screenshots for each visual component in the allocated slots.

6.1 Executive Summary Key Performance Indicators (Cards)



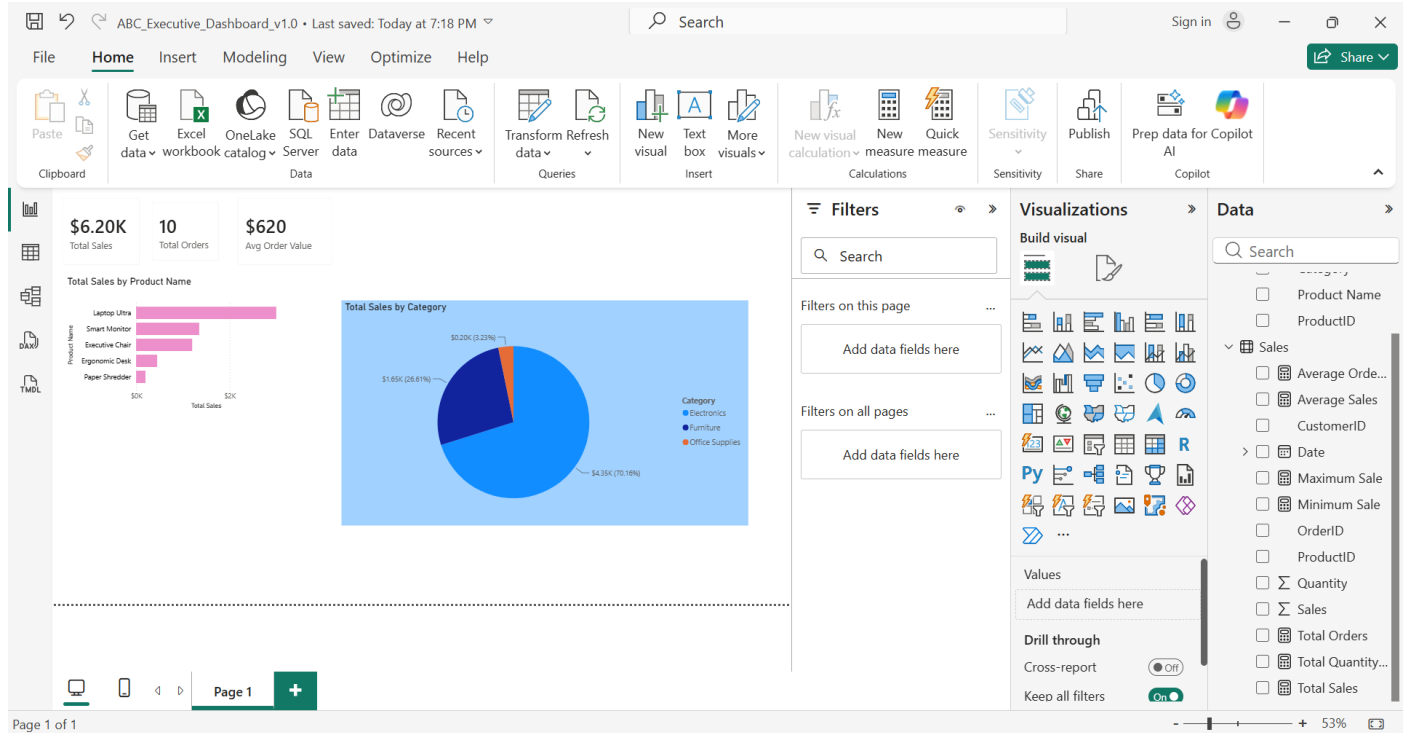
6.2 Core Analytical Visualizations

Clustered Bar Chart Sales by Product Name



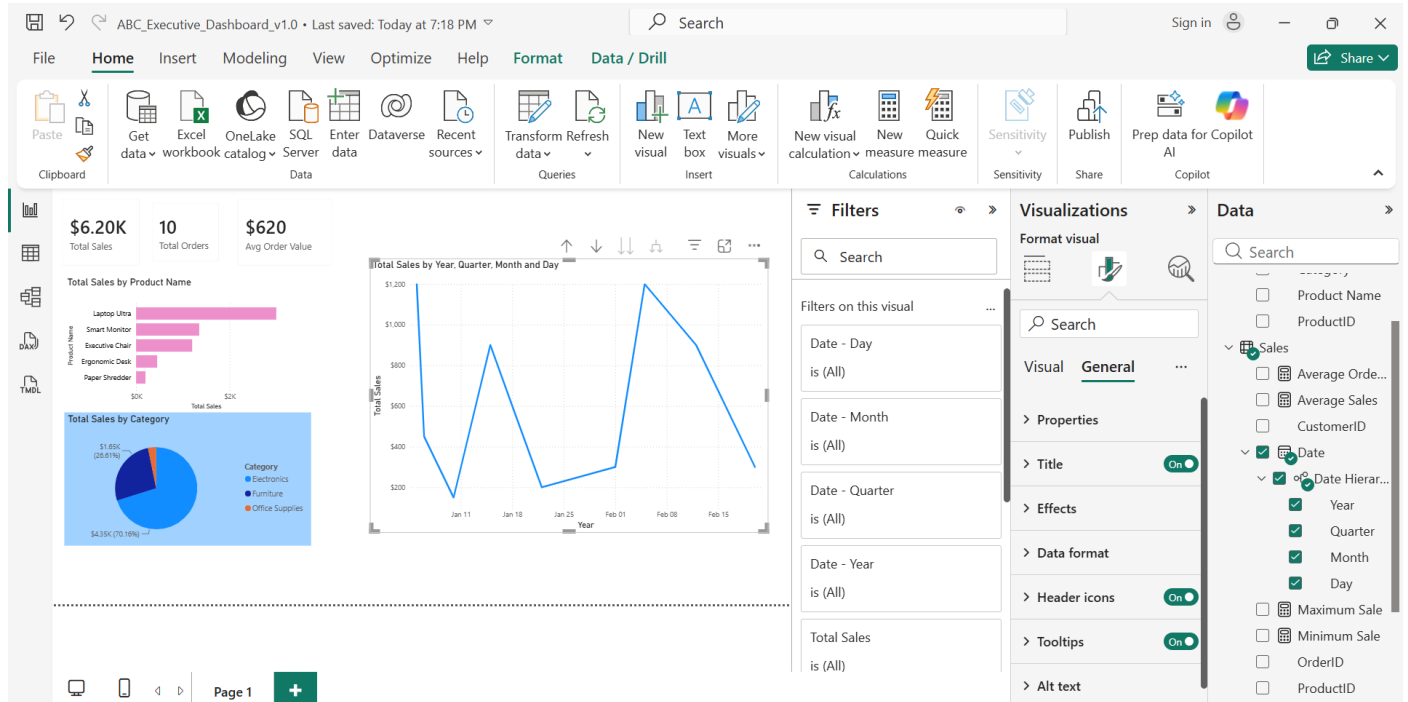
Pie Chart

Sales by Product Category



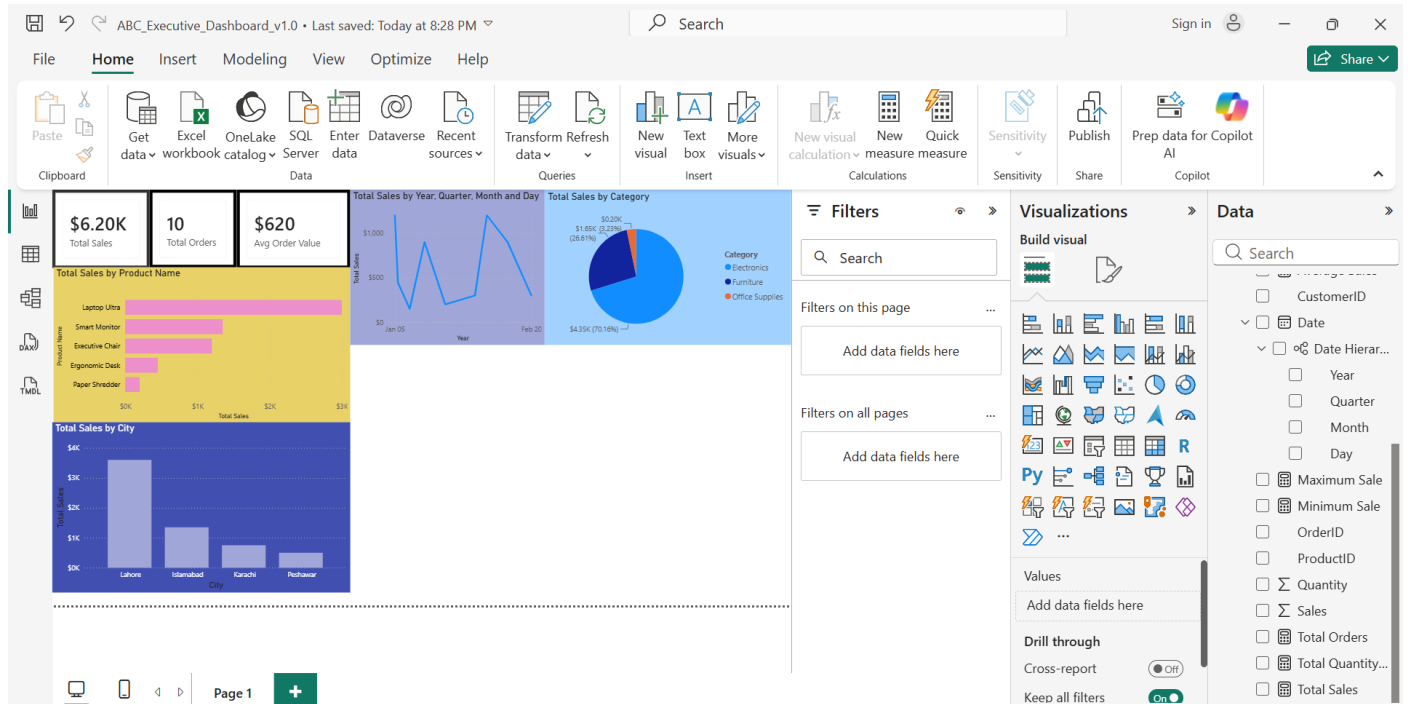
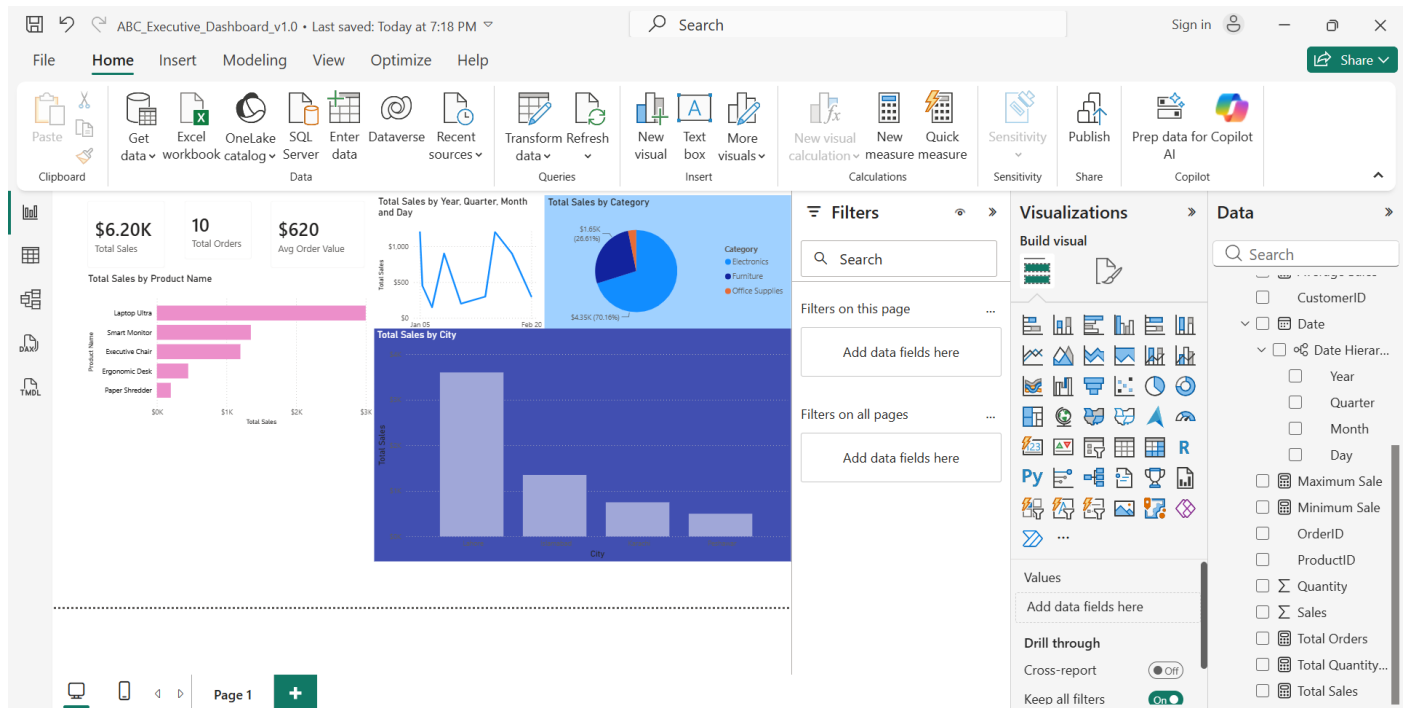
Line Chart

Sales Trend Over Time (Date)



Column Chart

Sales by Customer City



TASK 6: EXECUTIVE DASHBOARD VISUALS (CONT.)

6.3 Category Breakdown & Tabular Reports

Donut Chart

Revenue Share by Category

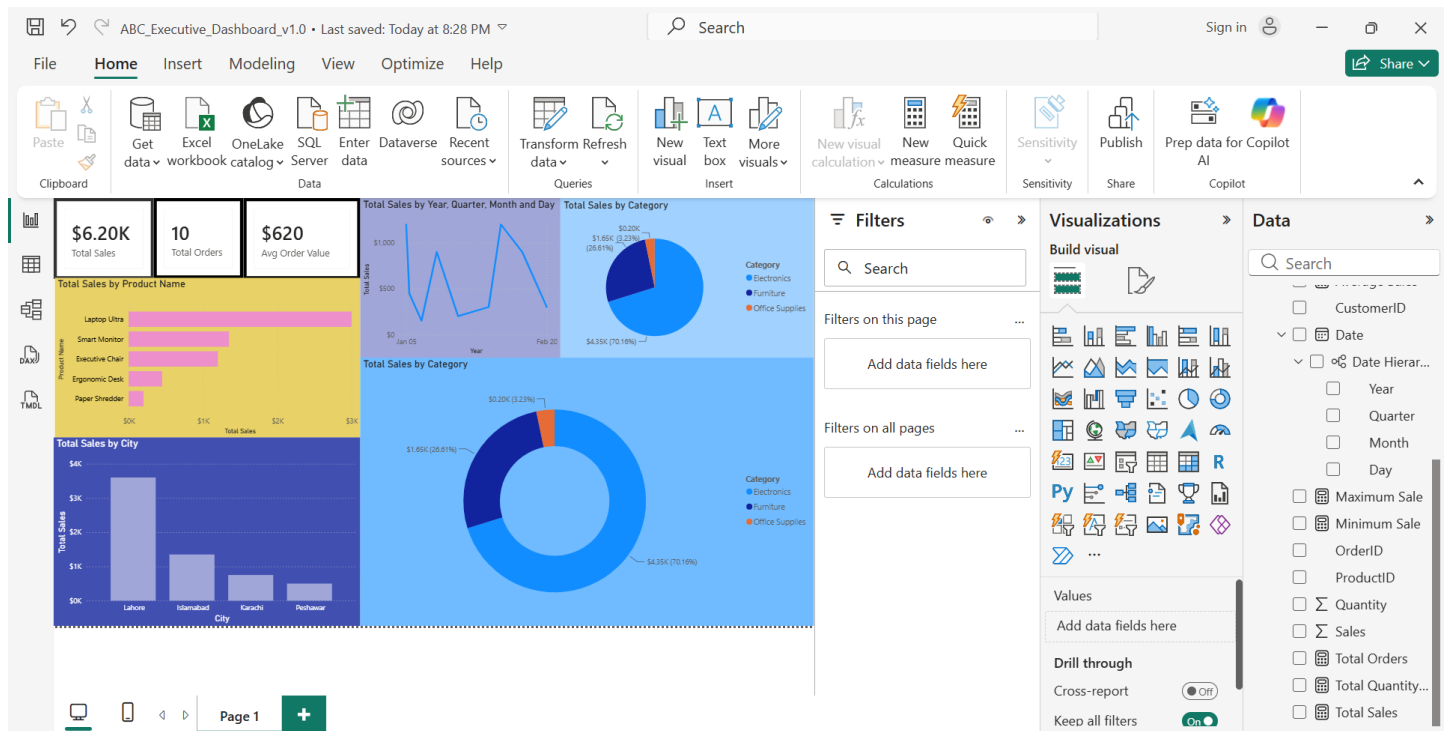


Table Visual

Customer-wise Sales Breakdown

Customer Name	Total Sales
Ali Khan	\$2,100.00
Ayesha Raza	\$1,500.00
Bilal Sheikh	\$500.00
Sara Ahmed	\$750.00
Usman Malik	\$1,350.00
Total	\$6,200.00

Matrix Visual

Category (Rows) vs City (Columns) with Values = Sum of Sales

Customer Name	Total Sales				
Ali Khan	\$2,100.00				
Ayesha Raza	\$1,500.00				
Bilal Sheikh	\$500.00				
Sara Ahmed	\$750.00				
Usman Malik	\$1,350.00				
Total	\$6,200.00				

Category	Islamabad	Karachi	Lahore	Peshawar	Total
Electronics	\$1,200.00	\$450.00	\$2,700.00		\$4,350.00
Furniture	\$150.00	\$300.00	\$900.00	\$300.00	\$1,650.00
Office Supplies			\$200.00	\$200.00	\$200.00
Total	\$1,350.00	\$750.00	\$3,600.00	\$500.00	\$6,200.00

6.4 Interactive Filters & KPI Targets

Slicer: Category

Dropdown / List



Slicer: City

Multi-select Slicer



Date Slicer

Between Date Range



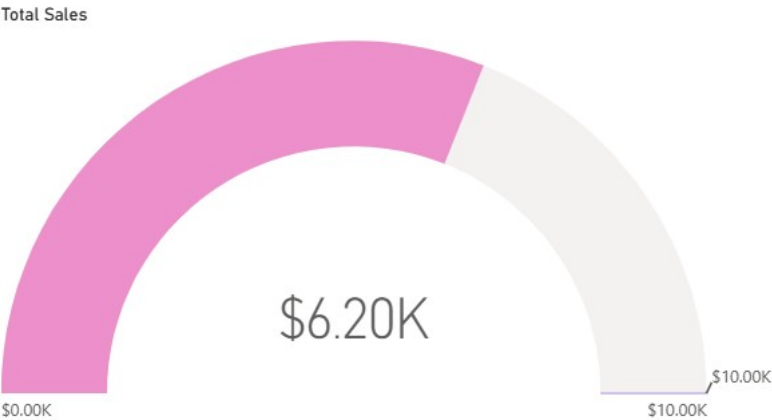
KPI Visual

Monthly Revenue Trend vs Goal



Gauge Chart

Sales Target Achievement (Current Sales vs \$10,000 Target)



TASK 7: DAX (DATA ANALYSIS EXPRESSIONS) ENGINEERING

10 MARKS

Instructions: Create an explicit DAX Measure Table (_Measures). Write the following explicit DAX expressions and attach screenshots of your DAX formula bar for each.

Total Sales

Total Sales = SUM(Sales[Sales])

 1 Total Sales = SUM(Sales[Sales])

Total Orders

Total Orders = DISTINCTCOUNT(Sales[OrderID])

 1 Total Orders = DISTINCTCOUNT(Sales[OrderID])

Average Order Value

Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)

 1 Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)

Total Quantity Sold

Total Quantity Sold = SUM(Sales[Quantity])

 1 Total Quantity Sold = SUM(Sales[Quantity])

Maximum Sale

Maximum Sale = MAX(Sales[Sales])

 1 Maximum Sale = MAX(Sales[Sales])

Minimum Sale

Minimum Sale = MIN(Sales[Sales])

 1 Minimum Sale = MIN(Sales[Sales])

Average Sales per Line Item

Average Sales = AVERAGE(Sales[Sales])

 1 Average Sales = AVERAGE(Sales[Sales])

TASK 8: BUSINESS DECISION INTELLIGENCE

10 MARKS

Instructions: Analyze the completed dashboard and answer the executive questions below based on the underlying datasets.

8.1 Analytical Questionnaire:

1. Which city generated the highest total revenue?

Answer: Lahore

2. Which specific product generated the highest revenue?

Answer: Laptop Ultra

3. Which product category performed best overall?

Answer: Electronics

4. What is the total order volume recorded across the period?

Answer: 10 total orders

5. What is the computed Average Order Value (AOV)?

Answer: \$620

8.2 Strategic Business Recommendations for C-Suite Leadership

Provide at least 5 high-impact strategic business recommendations for the CEO of ABC Retail Store based on your analytical findings.

Based on analysis of the ABC Retail Store dashboard (Total Sales: \$6,200 vs. \$10,000 goal — 62% attainment), the following strategic recommendations are presented for executive consideration:

1. Double Down on Electronics While Managing Concentration Risk Electronics is the clear revenue driver, generating \$4,350 (70% of total sales), led by Laptop Ultra (\$3,000) and Smart Monitor (\$1,350). Leadership should expand the Electronics assortment — accessories, peripherals, complementary SKUs — to capture additional wallet share. At the same time, this heavy dependency on one category represents a risk; a supply disruption or demand shift in Electronics would materially impact overall revenue, so diversification efforts should run in parallel.

2. Investigate and Revive Underperforming Categories Office Supplies contributes just \$200 (3% of sales) and Furniture only \$1,650 (27%). Leadership should determine whether this reflects genuine low demand, poor product visibility, weak pricing, or insufficient marketing investment. Depending on findings, either reinvest in promotion and merchandising for these categories or reallocate resources toward higher-performing lines.

3. Launch Targeted Regional Growth Campaigns Lahore generates \$3,600 (58% of total sales), more than four times Karachi's \$750 and over seven times Peshawar's \$500. This signals significant untapped potential outside the current stronghold. Recommend city-specific marketing campaigns, localized promotions, or regional sales incentives to grow underperforming markets rather than continuing to concentrate investment in Lahore alone.

4. Build a Customer Diversification Strategy Two customers — Ali Khan and Ayesha Raza — together account for \$3,600, or 58% of total revenue. This level of customer concentration is a material business risk. Leadership should invest in customer acquisition, loyalty programs, and referral incentives to broaden the revenue base and reduce dependency on a small number of high-value accounts.

5. Address the Declining Month-over-Month Sales Trend Sales fell from \$3,500 in January to \$2,700 in February — a 23% decline. This trend should be investigated immediately: possible causes include seasonality, stock-outs, or reduced marketing spend. Recommend establishing a monthly (or mid-month) sales review cadence, using the dashboard's KPI card and gauge visuals to catch downward trends earlier and trigger corrective action.

6. Reset and Phase the Path to the Sales Goal At only 62% of the \$10,000 target, the current goal is either unrealistic for the present trajectory or requires targeted intervention. Leadership should either set phased, milestone-based targets (by month, city, and category) or commit incremental investment — marketing spend, staffing, promotions — needed to close the 38% shortfall in a defined timeframe.

BONUS TASK: DASHBOARD UX/UI ENHANCEMENTS

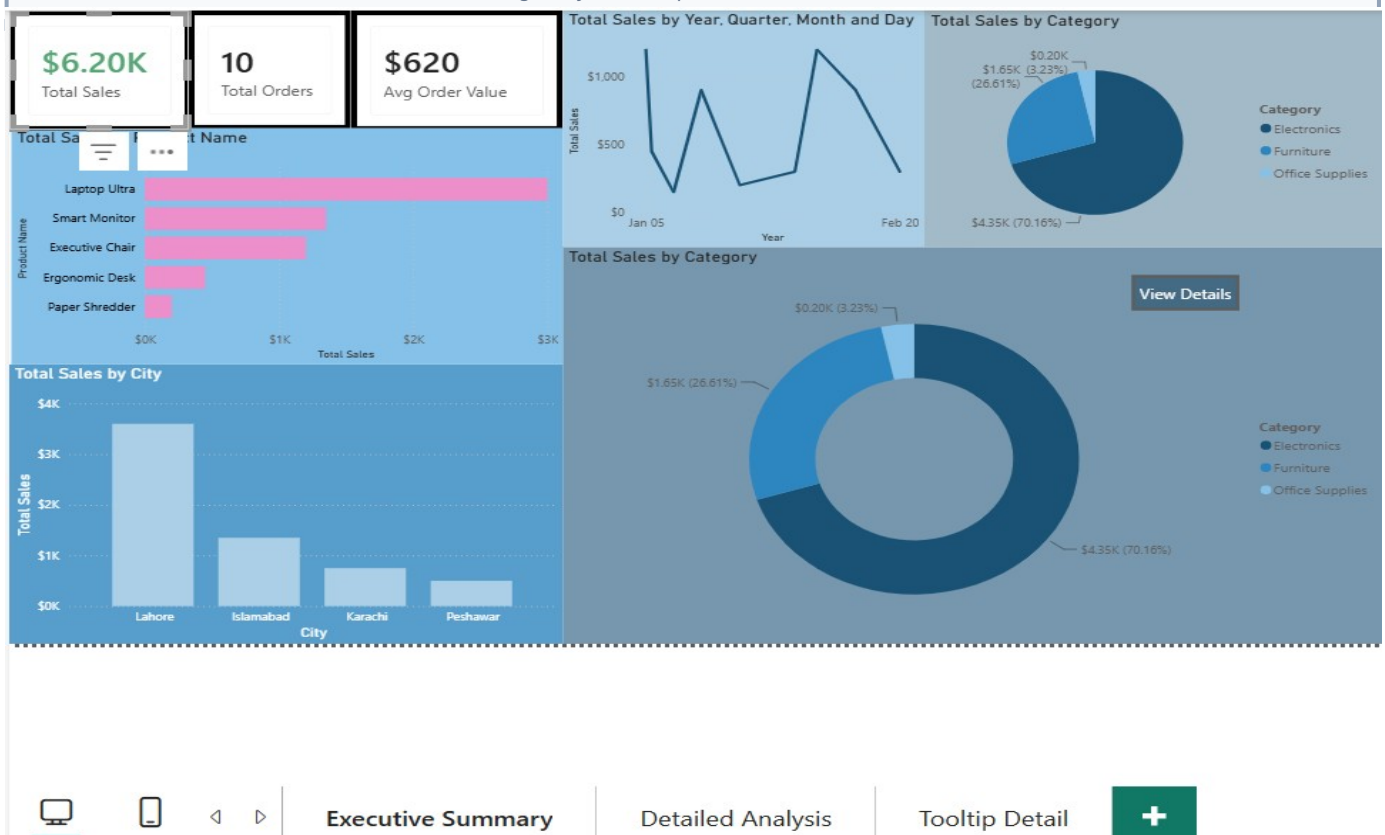
BONUS 10 MARKS

Incorporate modern dashboard UI features to transform the dashboard into an interactive analytical product:

- **Corporate Branding & Themes:** Custom blue theme palette matching company visual identity.
- **Interactive Navigation:** Page navigation buttons and dynamic Bookmark slicers.
- **Custom Tooltips:** Page-level report tooltips for micro-level detail on hover.
- **Conditional Formatting:** Color-scaled background fills on matrix cells and KPI cards.
- **Top-N Filtering:** Dynamic Top 3 Products filter options.

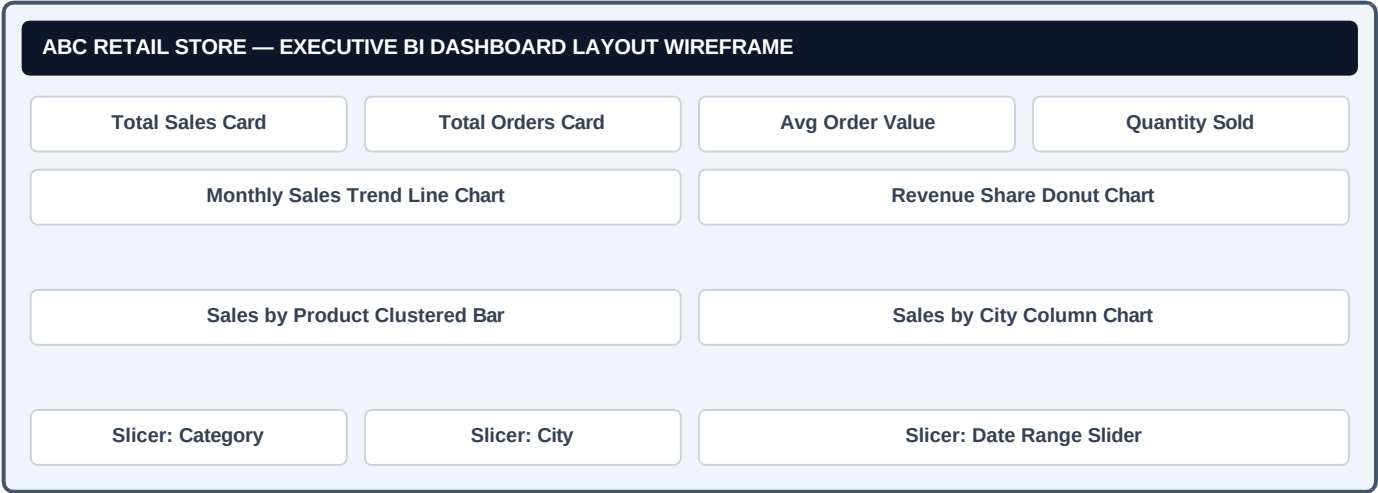
[SCREENSHOT PLACEHOLDER: ENHANCED EXECUTIVE DASHBOARD]

Paste full-screen image of your final polished Power BI Executive Dashboard.



Executive Dashboard Blueprint Wireframe

Use this reference layout when arranging dashboard visual elements in Power BI Desktop:



INDUSTRY BEST PRACTICES: ENTERPRISE BI STANDARDS

As a Data Engineer and BI Developer, adhering to enterprise standards ensures your dashboards are scalable, maintainable, and visually impactful.

1. Star Schema Modeling

Always design schemas with central Fact tables surrounded by single-direction Dimension tables. Avoid bi-directional filters and complex Snowflake relationships to optimize DAX engine performance.

2. Cognitive UI Design

Limit dashboards to 5-7 core visuals per page to prevent cognitive overload. Follow standard reading z-patterns (KPIs top, high-level trends middle, details bottom).

3. Color Harmony & Palette

Use a unified 3-color palette (e.g., Deep Navy primary, Slate secondary, Muted Blue accent). Reserve high-contrast colors (Red/Amber/Green) strictly for alert threshold KPIs.

4. Explicit DAX Measures

Never use implicit Power BI measures (drag-and-drop sums). Always write explicit DAX measures stored in dedicated measure tables for consistent calculations across reports.

Ten Commandments of Enterprise Dashboard Development:

- 1. Think C-Suite First:** Design for decision-makers who need answers within 5 seconds.
- 2. Ensure Visual Alignment:** Align all visuals pixel-perfectly using Power BI Format tools.
- 3. Dynamic Formatting:** Format all monetary values as Currency (\$) and quantities as Whole Numbers.
- 4. Remove Visual Clutter:** Disable unnecessary gridlines, background boxes, and dark borders.
- 5. Meaningful Chart Titles:** Use descriptive headers (e.g., "Revenue Trends by Quarter" instead of "Sales").
- 6. Optimize Performance:** Remove unreferenced columns in Power Query to shrink PBIX memory footprint.
- 7. Test Interactivity:** Verify cross-filtering behavior across all visual cards and slicers.
- 8. Mobile Layout Optimization:** Build responsive layouts for executive tablet and mobile views.
- 9. Version Control:** Maintain structured file naming conventions (e.g., ABC_Executive_Dashboard_v1.0.pbix).
- 10. Document Business Rules:** Comment complex DAX code and maintain data dictionaries.

EVALUATION RUBRIC & SUBMISSION CHECKLIST

Evaluation Criteria (100 Marks Total)

Evaluation Category	Marks	Assessment Criteria & Grading Standards
Understanding Concepts	15	Accurate definition of DW, OLTP vs OLAP comparison, and clear explanation of ETL phases.
Data Modeling	15	Correct classification of Fact and Dimension tables, identification of PK/FKs, and Star Schema design.
Relationships	10	Proper 1:* relationships established in Power BI with correct single-direction cross-filtering.
Dashboard Design	20	Professional UI layout, effective visual hierarchy, clear color scheme, and structured wireframe execution.
Visualizations	20	Implementation of required Cards, Bar, Line, Pie, Donut, Matrix, Slicers, and KPI visuals.
Business Insights	10	Accurate data interpretations and strategic recommendations tailored for executive leadership.
Presentation & Documentation	10	Complete documentation, clean screenshots, proper formatting, and adherence to submission guidelines.
Total Marks	100	Pass Threshold: 70% Distinction: 90%

Submission Checklist

Ensure all required project artifacts are finalized prior to submitting your assignment bundle:

☐ **Power BI File (.pbix):** Fully configured data model.

☐ **Excel Dataset (.xlsx):** Cleaned source data tables.

☐ **Star Schema Diagram:** Completed architecture drawing.

☐ **Data Relationships:** Verified Model View screenshots.

☐ **All Visualizations:** Embedded chart screenshots.

☐ **KPI Cards:** Configured Executive summary metrics.

☐ **DAX Measures:** Explicit code snippets provided.

☐ **Business Insights:** C-suite written recommendations.

☐ **Dashboard Screenshots:** Full UI layout images.

☐ **Final PDF Booklet:** Completed student report.

Submission Portal: Upload your PBIX and completed Assignment PDF to the University Learning Management System (LMS) under **Week 5 Data Engineering Portal** before the deadline.